Abdelrahman Khalifa

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Summary

Embedded software engineer who works upstream. Built custom Linux images with Yocto and Buildroot, wrote a Linux kernel module for ARM, and booted ARM boards in QEMU with U-Boot, device trees and BusyBox root file systems. Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting), with 11 PRs merged upstream in C and Rust, plus merged work in Zephyr RTOS. Owns the Zephyr firmware and firmware-update path for EKSON's motion simulators.

Experience

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Built the **Printers page for COSMIC Settings** and a standalone app for the Linux printing stack (CUPS, IPP, DNS-SD/Avahi, Printer Applications), worked remotely with mentors from OpenPrinting and System76 (cosmic-settings #2182, in review).
- Got **11 PRs merged upstream** (cups-rs, #164, #165): ported cups-rs to libcups3 with a GitHub Actions matrix across architectures and CUPS versions, added DNS-SD FFI bindings, and fixed crashes and a leaked client in libcups' Avahi code (C).

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS; PID valve control with IMU feedback, safety stops and a watchdog.
- Built remote operation and firmware updates: the desktop app polls a Node.js/MongoDB backend over REST and flashes new firmware with esptool; GitHub Actions CI.

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Created custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; scored 100% in the final evaluation.

OSS Projects

Zephyr RTOS Mar 2025

- Implemented the `IP_MULTICAST_LOOP` socket option for IPv4 across 6 networking modules, reviewed and merged (issue #86977).

Projects

Kernel Monitor | *C, Linux kernel, ARM* GitHub

- Character device driver that walks the process list under RCU and exports per-task memory, threads and CPU time; run on QEMU vexpress-a9 with a custom kernel, device tree and BusyBox ext4 root file system.

A/B firmware updates | *graduation project; STM32, ESP32* GitHub

- An ESP32 gateway sends the image over UART with UDS services; the STM32 bootloader verifies it with the hardware CRC unit and uses two banks with rollback.

Technical Skills

Embedded Linux: Yocto, Buildroot, U-Boot, Linux kernel modules and drivers, device trees, BusyBox root file systems, QEMU (ARM), cross-compilation, Raspberry Pi
Languages: C, Rust, Modern C++, Python, Bash
Linux & tools: Debian/Ubuntu, Arch (CachyOS), CUPS/IPP, DNS-SD/Avahi, varlink/D-Bus IPC; Git, GitHub Actions CI, CMake, Make, Cargo, upstream code review

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors. Codeforces Expert; 4-time ECPC finalist (2021–2024).